```
#include<iostream>
using namespace std;
int linear_search ( int a[],int n, int x )
{
   for (int i=0; i<n; i++)
   {
      if (a[i]==x)
      return(i);

   }
   return (-1);

}
int main (){
   int n;
   int a[20];
   cout<<"enter the size of the array:";
   cin >>n;
   cout<<"enter"<<n<<"integers:";
   for (int i=0; i<n; i++){
      cin >>a[i];

   }
   int x;
   cout<<"enter the target element to search for:";
   cin >>x;
int location=linear_search (a,n,x);
if (location!=-1){
   cout<<"element"<<x<<"found at index:"<<location;}
   else{
      cout<<"element"<< x<<"not found in array";
   }
   return 0;
}

/*
PS D:\swarupa44> cd "d:\swarupa44\" ; if ($?) { g++ linearsearch.cpp -o linearsearch } ;
if
($?) { .\linearsearch }
enter the size of the array:5
enter5integers:10
20
30
40
50
```

enter the target element to search for:40
element40found at index:3
*/